

MALAVIKA KASHIREDDY

Chicago, IL | kashireddymalavika@gmail.com | (331) 285-3241 | linkedin.com/in/malavika-kashireddy | github.com/malavikakashireddy

EDUCATION

University of Illinois Chicago — Chicago, IL

Bachelor of Science in Computer Science | Expected Dec 2026

Relevant Coursework: Algorithms, Data Structures, Applied Statistical Methods (R), Discrete Mathematics, Linear Algebra, Computer Architecture, Machine Organization, Programming Practicum

EXPERIENCE

UI Health — Chicago, IL

Software Engineer Intern | May 2025 – Jul 2025

- Partnered with engineering, research, and physician stakeholders to scope, design, and launch a metrics and observability MVP for **1-Search**, a **physician-facing healthcare platform serving 7,000+ users**.
- Translated stakeholder needs into technical requirements and shipped **5 production Django REST API endpoints** plus centralized application-event logging, enabling real-time visibility into search performance and user behavior.
- Designed analytics data flows capturing search duration, results, and application events, giving research teams measurable signals to evaluate product performance and clinical search workflows.
- Delivered production backend capabilities that remain in active use, supporting ongoing analysis of platform performance and user interaction data.

Discovery Partners Institute — Chicago, IL

Student Worker, R&D Support | Feb 2024 – Dec 2025

- Productized progress-tracking dashboards for **HERO and Science Teams**, centralizing research, funding, and project metrics for a **10–15 person R&D organization**.
- Consolidated data across **6 research institutions**—Argonne, UIUC, Northwestern, Cardiff, UIS, and UIC—into at-a-glance views of budgets, proposal pipelines, researcher activity, and collaborations.
- Managed data for **2 research initiatives**, tracking grant budgets, proposal status, researcher contributions, and project progress to improve visibility across concurrent workflows.
- Supported cross-functional research operations through workflow automation, technical support, documentation, and data management across multiple teams.

MSLC, University of Illinois Chicago — Chicago, IL

Learning Assistant | Oct 2024 – Dec 2024

- Supported **50+ students per session**, adapting complex technical concepts and problem-solving approaches to users with different backgrounds and levels of understanding.

PRODUCT & TECHNICAL PROJECTS

DevDocs AI | Python, FastAPI, PostgreSQL, pgvector, React, Docker

- Designed an AI-powered developer knowledge product to reduce friction in navigating fragmented source code and technical documentation using **RAG, hybrid retrieval, embeddings, and LLM-generated answers with source references**.
- Built an end-to-end ingestion and retrieval pipeline across **1,000+ files and ~5,000 indexed chunks**, spanning code parsing, chunking, embeddings, metadata indexing, and semantic retrieval.
- Defined and measured retrieval quality against a **250-query evaluation benchmark**, using hybrid search and reranking to achieve **~85% Recall@5** and iteratively improve answer relevance.

Real-Time Analytics & Observability Platform | Go, Apache Kafka, PostgreSQL, Redis, React, Grafana, Docker

- Built a real-time analytics product that transforms application-event data into actionable visibility across **traffic, latency, errors, and user activity**.
- Designed an event-driven pipeline sustaining **5,000+ events/sec** with **sub-2-second aggregation latency**, processing **500,000+ events** across 6 concurrent consumers.
- Developed React and Grafana dashboards for real-time product and system metrics, maintaining **99%+ event-processing success** under sustained load testing.

Distributed Job Processing System | Python, FastAPI, PostgreSQL, Redis, Docker

- Designed a distributed task-processing platform for reliable asynchronous execution, incorporating scheduling, job states, retries, failure handling, and worker coordination.
- Processed **2,000+ tasks across 4 workers** and achieved **~3× throughput improvement to 150 jobs/sec** through concurrent execution.

LEADERSHIP & USER ENGAGEMENT

Shorelight (UIC Global) | Student Ambassador, User Research & Outreach | Feb 2026 – Present

- Conduct **1:1 discovery conversations** with prospective international students, identifying questions, friction points, and needs throughout the university decision journey.
- Synthesize qualitative feedback into actionable insights for admissions and marketing teams while supporting webinars, campus tours, and student-facing programs.

University of Illinois Chicago | Orientation Leader | Fall 2026

- Supported **250 incoming students** through their campus transition, serving as a first point of contact and communicating complex university resources to a diverse user population.

SKILLS

Product & Analytics: User Discovery, User Research, Requirements Gathering, Product Analytics, MVP Scoping, Success Metrics, Cross-Functional Collaboration, Stakeholder Communication, Data Analysis, Data Visualization, Agile

Technical: Python, Java, C++, C, Go, TypeScript, JavaScript, SQL, Django REST Framework, FastAPI, PostgreSQL, Redis, Kafka, React, Grafana, Docker, AWS, Git, GitHub Actions

AI/ML: LLMs, RAG, Embeddings, Vector Search, pgvector, OpenAI API, LangChain, Gemini